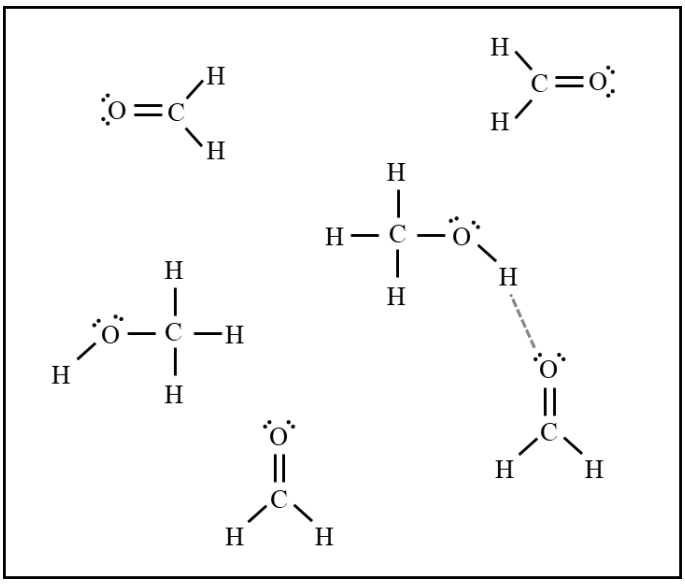
Question 4: Short Answer — 4 points

A For the correct answer: **Point 01**

sp^2

B For a correct diagram: **Point 02**

The diagram should show a dashed line between the O *atom in one* H_2CO *molecule and the* H *atom in the* –OH *group of one* CH_3OH *molecule. See example response below.*



C (i) For a correct proposal: **Point 03**

The proposed temperature should be in the range 181 K – 254 K .

(ii) For the correct calculated value: **Point 04**

$$8.59 \text{ g } CH_3OH \times \frac{1 \text{ mol } CH_3OH}{32.04 \text{ g } CH_3OH} \times \frac{-37.6 \text{ kJ}}{1 \text{ mol } CH_3OH} = -10.1 \text{ kJ}, \textit{ so } 10.1 \text{ kJ } \textit{are removed.}$$

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